SG No. 192 D_{6h}^2 $P6/mcc$ [hexagonal]

* plus set: $+[0,\ 0,\ 0]$

Table 1: Wyckoff site: 2a, site symmetry: 622

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12]
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	[13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: 2b, site symmetry: 6/m..

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,2,3,4,5,6,13,14,15,16,17,18]
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	[7,8,9,10,11,12,19,20,21,22,23,24]

Table 3: Wyckoff site: 4c, site symmetry: 3.2

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	[1,2,3,10,11,12]
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	[4,5,6,7,8,9]
3	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	[13,14,15,22,23,24]
4	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[16,17,18,19,20,21]

Table 4: Wyckoff site: 4d, site symmetry: -6..

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	[1,2,3,16,17,18]
2	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	[4,5,6,13,14,15]
3	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[7,8,9,22,23,24]
4	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	[10,11,12,19,20,21]

Table 5: Wyckoff site: 4e, site symmetry: 6..

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,2,3,4,5,6]
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[7,8,9,10,11,12]
3	$[0,\ 0,\ -z]$	[13,14,15,16,17,18]
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[19,20,21,22,23,24]

Table 6: Wyckoff site: 6f, site symmetry: 222

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[1,4,8,11]
2	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	[2,5,7,10]

continued ...

Table 6

No.	position	mapping
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[3,6,9,12]`
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[13,16,20,23]`
5	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[14,17,19,22]`
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[15,18,21,24]`

Table 7: Wyckoff site: `6g`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,13,16]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,5,14,17]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,6,15,18]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[7,10,19,22]`
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,11,20,23]`
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,12,21,24]`

Table 8: Wyckoff site: `8h`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,2,3]`
2	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[4,5,6]`
3	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2} - z\right]$	`[7,8,9]`
4	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	`[10,11,12]`
5	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[13,14,15]`
6	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[16,17,18]`
7	$\left[\frac{1}{3}, \frac{2}{3}, z + \frac{1}{2}\right]$	`[19,20,21]`
8	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	`[22,23,24]`

Table 9: Wyckoff site: `12i`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,4]`
2	$\left[0, \frac{1}{2}, z\right]$	`[2,5]`
3	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3,6]`
4	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	`[7,10]`
5	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	`[8,11]`
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[9,12]`
7	$\left[\frac{1}{2}, 0, -z\right]$	`[13,16]`
8	$\left[0, \frac{1}{2}, -z\right]$	`[14,17]`
9	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[15,18]`
10	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	`[19,22]`
11	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	`[20,23]`
12	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[21,24]`

Table 10: Wyckoff site: `12j`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,8]`
2	$\left[0,\ x,\ \frac{1}{4}\right]$	`[2,7]`
3	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[3,9]`
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,11]`
5	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[5,10]`
6	$\left[x,\ x,\ \frac{1}{4}\right]$	`[6,12]`
7	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[13,20]`
8	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[14,19]`
9	$\left[x,\ x,\ \frac{3}{4}\right]$	`[15,21]`
10	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[16,23]`
11	$\left[0,\ x,\ \frac{3}{4}\right]$	`[17,22]`
12	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[18,24]`

Table 11: Wyckoff site: `12k`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,11]`
2	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[2,10]`
3	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[3,12]`
4	$\left[-x,\ -2x,\ \frac{1}{4}\right]$	`[4,8]`
5	$\left[2x,\ x,\ \frac{1}{4}\right]$	`[5,7]`
6	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[6,9]`
7	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[13,23]`
8	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[14,22]`
9	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[15,24]`
10	$\left[x,\ 2x,\ \frac{3}{4}\right]$	`[16,20]`
11	$\left[-2x,\ -x,\ \frac{3}{4}\right]$	`[17,19]`
12	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[18,21]`

Table 12: Wyckoff site: `12l`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,16]`
2	$[-y,\ x-y,\ 0]$	`[2,17]`
3	$[-x+y,\ -x,\ 0]$	`[3,18]`
4	$[-x,\ -y,\ 0]$	`[4,13]`
5	$[y,\ -x+y,\ 0]$	`[5,14]`
6	$[x-y,\ x,\ 0]$	`[6,15]`
7	$\left[y,\ x,\ \frac{1}{2}\right]$	`[7,22]`
8	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	`[8,23]`
9	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	`[9,24]`
10	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[10,19]`
11	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[11,20]`
12	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[12,21]`

Table 13: Wyckoff site: `24m`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-y,\, x-y,\, z]$	`[2]`
3	$[-x+y,\, -x,\, z]$	`[3]`
4	$[-x,\, -y,\, z]$	`[4]`
5	$[y,\, -x+y,\, z]$	`[5]`
6	$[x-y,\, x,\, z]$	`[6]`
7	$\left[y,\, x,\, \frac{1}{2}-z\right]$	`[7]`
8	$\left[x-y,\, -y,\, \frac{1}{2}-z\right]$	`[8]`
9	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	`[9]`
10	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	`[10]`
11	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	`[11]`
12	$\left[x,\, x-y,\, \frac{1}{2}-z\right]$	`[12]`
13	$[-x,\, -y,\, -z]$	`[13]`
14	$[y,\, -x+y,\, -z]$	`[14]`
15	$[x-y,\, x,\, -z]$	`[15]`
16	$[x,\, y,\, -z]$	`[16]`
17	$[-y,\, x-y,\, -z]$	`[17]`
18	$[-x+y,\, -x,\, -z]$	`[18]`
19	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	`[19]`
20	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	`[20]`
21	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	`[21]`
22	$\left[y,\, x,\, z+\frac{1}{2}\right]$	`[22]`
23	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	`[23]`
24	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	`[24]`